



COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

Computer Networks and the Internet

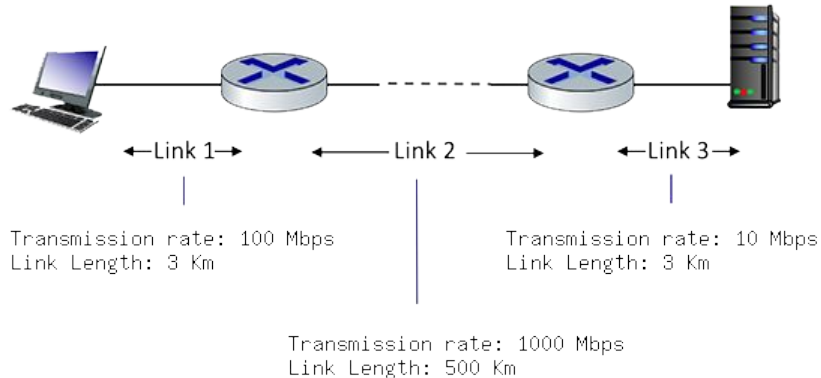
Interactive Exercises

Team Networks

Department of Computer Science and Engineering

Computing end-end delay (transmission and propagation delay)

Consider the figure below, with three links, each with the specified transmission rate and link length.



Find the end-to-end delay (including the transmission delays and propagation delays on each of the three links, but ignoring queueing delays and processing delays) from when the left host begins transmitting the first bit of a packet to the time when the last bit of that packet is received at the server at the right.

The speed of light propagation delay on each link is 3×10^8 m/sec.

Note that the transmission rates are in Mbps and the link distances are in Km.

Assume a packet length of 8000 bits. Give your answer in milliseconds.

Solution:

Link 1 transmission delay = $L/R = 8000 \text{ bits} / 100 \text{ Mbps} = 80 \text{ microsec}$

Link 1 propagation delay = $d/s = 3 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 10 \text{ microsec}$

Link 1 total delay = $80 + 10 = 90 \text{ microsec}$

Link 2 transmission delay = $L/R = 8000 \text{ bits} / 1000 \text{ Mbps} = 8 \text{ microsec}$

Link 2 propagation delay = $d/s = 500 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 1670 \text{ microsec}$

Link 2 total delay = $8 + 1670 = 1678 \text{ microsec}$

Link 3 transmission delay = $L/R = 8000 \text{ bits} / 10 \text{ Mbps} = 800 \text{ microsec}$

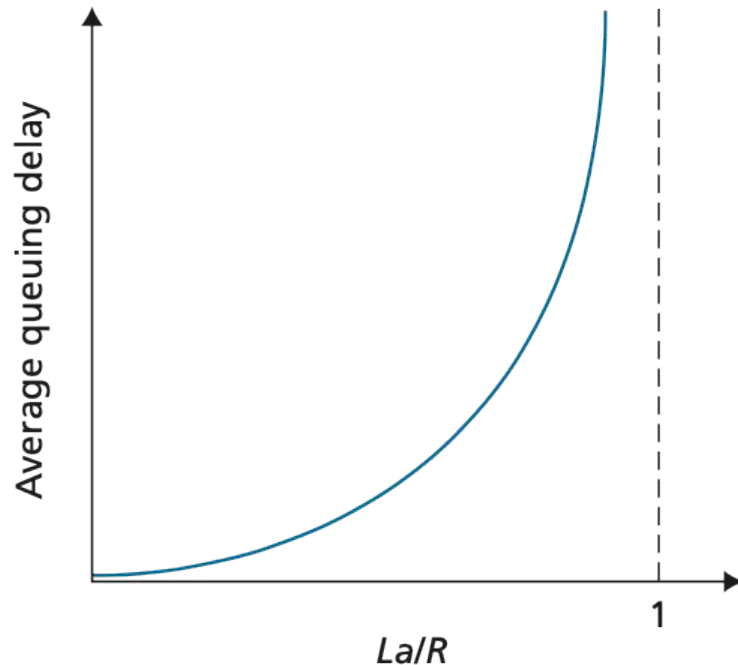
Link 3 propagation delay = $d/s = 3 \text{ Km} / 3 \times 10^8 \text{ m/sec} = 10 \text{ microsec}$

Link 3 total delay = $800 + 10 = 810 \text{ microsec}$

Total end-to-end delay is the sum of these six delays: $90 + 1678 + 810 = 2.578 \text{ ms}$

Computing Queuing Delay

Consider the queuing delay in a router buffer.



Assume:

a constant transmission rate of

$R = 1800000$ bps,

a constant packet-length $L = 7300$ bits, and

λ is the average rate of packets/second.

Traffic intensity $I = \lambda L / R$, and

Queuing delay is calculated as:

$$\frac{L}{R(1 - I)} \text{ for } I < 1.$$

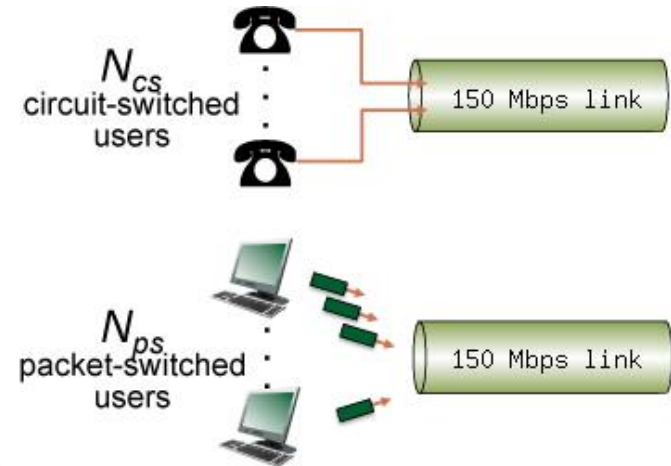
1. In practice, does the queuing delay tend to vary a lot? Answer with Yes or No
2. Assuming that $a = 30$, what is the queuing delay?
3. Assuming that $a = 76$, what is the queuing delay?
4. Assuming the router's buffer is infinite, the queuing delay is 0.8647 ms, and 1762 packets arrive. How many packets will be in the buffer 1 second later?
5. If the buffer has a maximum size of 956 packets, how many of the packets would be dropped upon arrival from the previous question?

1. Yes, in practice, queuing delay can vary significantly. We use the above formulas as a way to give a rough estimate, but in a real-life scenario it is much more complicated.
2. Queuing Delay = $\frac{IL}{R(1 - I)} * 1000 = \frac{0.1217 * 7300}{(1800000 * (1 - 0.1217))} * 1000 = 0.561 \text{ ms.}$
3. Queuing Delay = $\frac{IL}{R(1 - I)} * 1000 = \frac{0.3082 * 7300}{(1800000 * (1 - 0.3082))} * 1000 = 1.806 \text{ ms.}$
4. Packets left in buffer = $a - \text{floor}(1000/\text{delay}) = 1762 - \text{floor}(1000/0.8647) = 606 \text{ packets.}$
5. Packets dropped = packets - buffer size = $1762 - 956 = 806 \text{ dropped packets.}$

Quantitative Comparison of Packet Switching and Circuit Switching

Consider the two scenarios below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 10 Mbps, must share a link of capacity 150 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 150 Mbps link, where each user again requires 10 Mbps when transmitting, but only needs to transmit 30 percent of the time.



Round your answer to two decimals after leading zeros

Answer the following questions:

1. When circuit switching is used, what is the maximum number of circuit-switched users that can be supported? Explain your answer.
2. For rest of the questions, suppose packet switching is used. Suppose there are 29 packet-switching users (i.e., $N_{ps} = 29$). Can this many users be supported under circuit-switching? Explain.
3. When one user is transmitting, what fraction of the link capacity will be used by this user? Write your answer as a decimal.

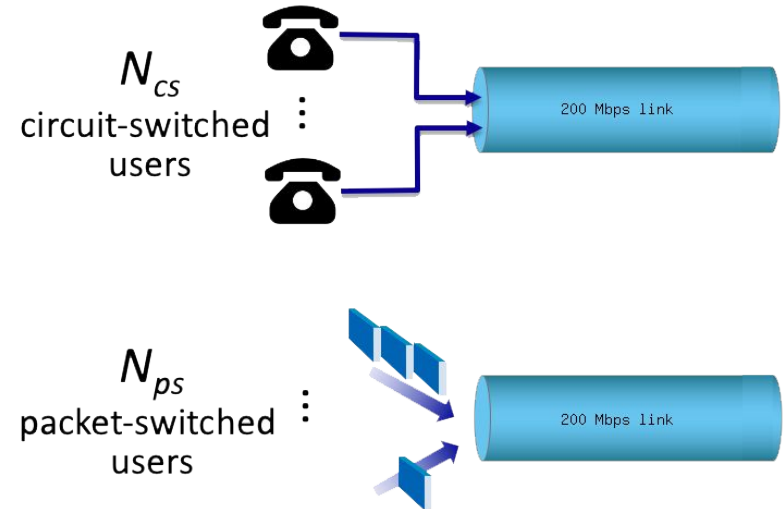
Answers:

1. 15
2. No.
3. This user will be transmitting at a rate of 10 Mbps over the 150 Mbps link, using a fraction $10/100 = \mathbf{0.06}$ of the link's capacity when busy.

Quantitative Comparison of Packet Switching and Circuit Switching

Consider the two scenarios below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 200 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 200 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 20 percent of the time.



Answer the following questions:

1. When circuit switching is used, what is the maximum number of users that can be supported?
2. Suppose packet switching is used. If there are 15 packet-switching users, can this many users be supported under circuit-switching? Yes or No.
3. When one user is transmitting, what fraction of the link capacity will be used by this user? Write your answer as a decimal.

Answers:

1. 8
2. No.
3. This user will be transmitting at a rate of 25 Mbps over the 200 Mbps link, using a fraction $25/200 = \mathbf{0.125}$ of the link's capacity when busy.

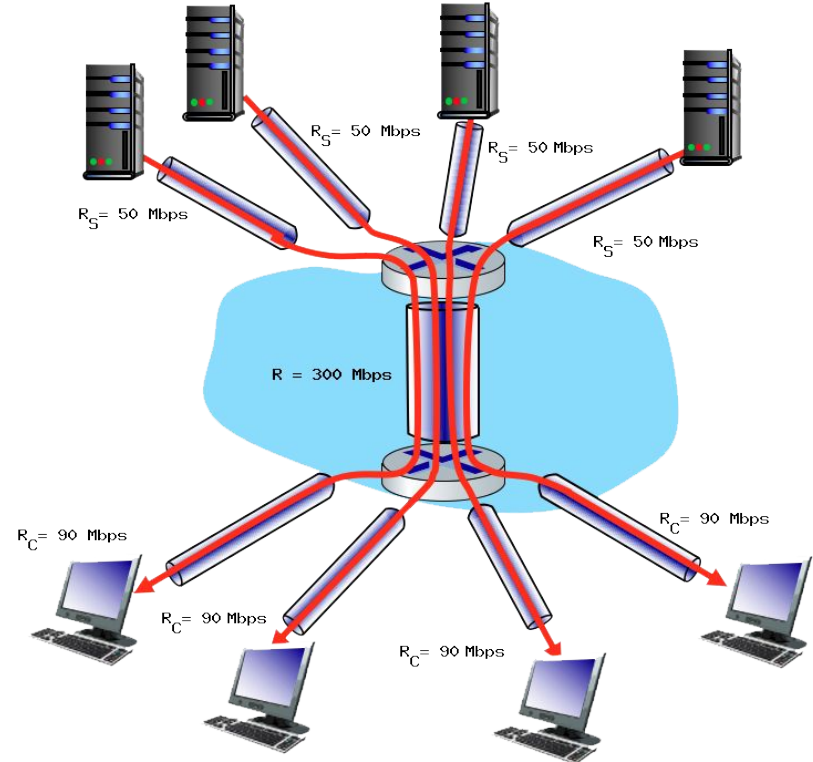
END TO END THROUGHPUT AND BOTTLENECK LINKS

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths.

The four pairs share a common middle hop with a transmission capacity of $R = 300$ Mbps.

The four links from the servers to the shared link have a transmission capacity of $R_S = 50$ Mbps.

Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 90$ Mbps.



Answer the following questions:

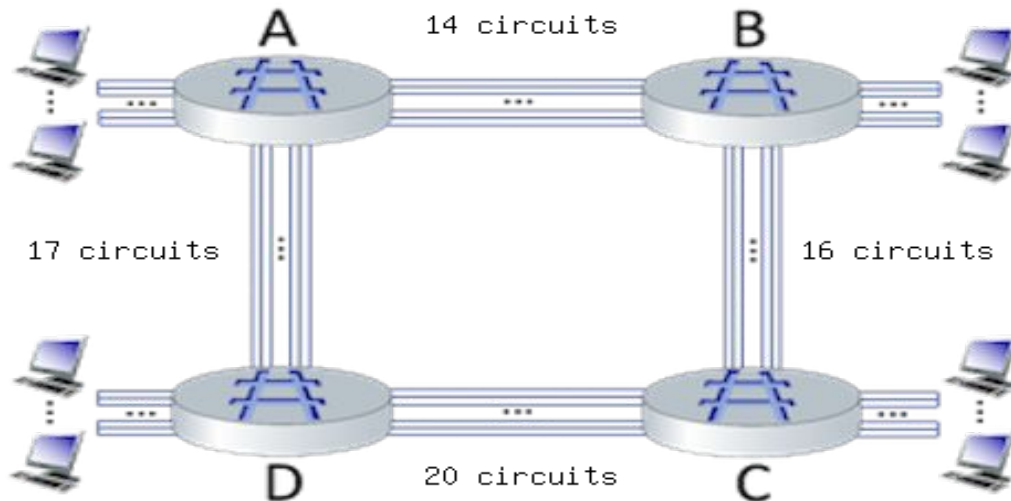
1. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?
2. Which link is the bottleneck link? Format as R_c , R_s , or R
3. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_s)? Answer as a decimal
4. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_c)? Answer as a decimal
5. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal

Solutions:

1. The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 50 Mbps
2. The bottleneck link is the link with the smallest capacity between R_S , R_C , and $R/4$. The bottleneck link is R_S .
3. The server's utilization = $R_{\text{bottleneck}} / R_S = 50 / 50 = 1$
4. The client's utilization = $R_{\text{bottleneck}} / R_C = 50 / 90 = 0.56$
5. The shared link's utilization = $R_{\text{bottleneck}} / (R / 4) = 50 / (300 / 4) = 0.67$

Packet Switching vs Circuit Switching – Numerical Example

Consider the circuit-switched network shown in the figure below, with circuit switches A, B, C, and D. Suppose there are 14 circuits between A and B, 16 circuits between B and C, 20 circuits between C and D, and 17 circuits between D and A



Question List

1. What is the maximum number of connections that can be ongoing in the network at any one time?
2. Suppose that these maximum number of connections are all ongoing. What happens when another call connection request arrives to the network, will it be accepted? Answer Yes or No
3. Suppose that every connection requires 2 consecutive hops, and calls are connected clockwise. For example, a connection can go from A to C, from B to D, from C to A, and from D to B. With these constraints, what is the maximum number of connections that can be ongoing in the network at any one time?
4. Suppose that 12 connections are needed from A to C, and 15 connections are needed from B to D. Can we route these calls through the four links to accommodate all 27 connections? Answer Yes or No



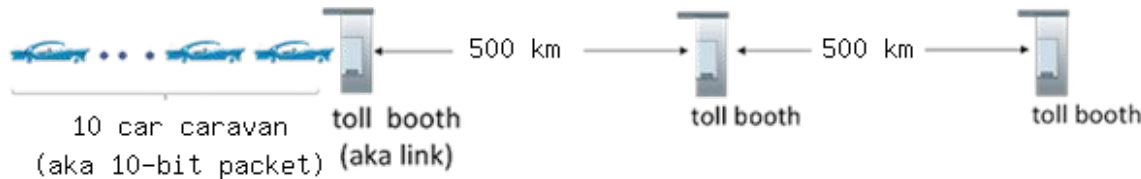
Solution

1. The maximum number of connections that can be ongoing at any one time is the sum of all circuits, which happens when 14 connections go from A to B, 16 connections go from B to C, 20 connections go from C to D, and 17 connections go from D to A. This sum is 67.
2. No, it will be blocked because there are no free circuits.
3. There can be a maximum of 33 connections. Consider routes A->C and C->A, sum the bottleneck links, consider any leftover capacity that would allow for B->D and D->B connections, and compare that value to the equivalent of B->D and D->B.
4. Using our answer from question 4, the sum of our needed connections is 27, and we have 33 available connections, so it is possible.



Car - Caravan Analogy

Consider the figure below, adapted from Figure 1.17 in the text, which draws the analogy between store-and-forward link transmission and propagation of bits in packet along a link, and cars in a caravan being serviced at a toll booth and then driving along a road to the next toll booth.



Suppose the caravan has 10 cars, and that the tollbooth services (that is, transmits) a car at a rate of one car per 2 seconds. Once receiving serving a car proceeds to the next toll booth, which is 500 kilometers away at a rate of 10 kilometers per second. Also assume that whenever the first car of the caravan arrives at a tollbooth, it must wait at the entrance to the tollbooth until all of the other cars in its caravan have arrived, and lined up behind it before being serviced at the toll booth. (That is, the entire caravan must be stored at the tollbooth before the first car in the caravan can pay its toll and begin driving towards the next tollbooth).



Question List

1. Once a car enters service at the tollbooth, how long does it take until it leaves service?
2. How long does it take for the entire caravan to receive service at the tollbooth (that is the time from when the first car enters service until the last car leaves the tollbooth)?
3. Once the first car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?
4. Once the last car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?
5. Once the first car leaves the tollbooth, how long does it take until it enters service at the next tollbooth?
6. Are there ever two cars in service at the same time, one at the first toll booth and one at the second toll booth?
Answer Yes or No
7. Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first toll booth but not yet arrived at the second tollbooth? Answer Yes or No



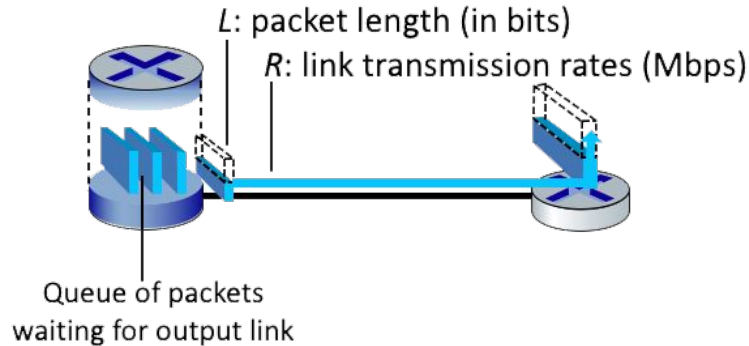
Solution

1. Service time is 2 seconds
2. It takes 20 seconds to service every car, (10 cars * 2 seconds per car)
3. It takes 50 seconds to travel to the next toll booth (500 km / 10 km/s)
4. Just like in the previous question, it takes 50 seconds, regardless of the car
5. It takes 68 seconds until the first car gets serviced at the next toll booth (10-1 cars * 2 seconds per car + 500 km / 10 km/s)
6. No, because cars can't get service at the next tollbooth until all cars have arrived
7. Yes, one notable example is when the last car in the caravan is serviced but is still travelling to the next toll booth; all other cars have to wait until it arrives, thus no cars are being serviced



Computing the one-hop transmission delay

Consider the figure below, in which a single router is transmitting packets, each of length L bits, over a single link with transmission rate R Mbps to another router at the other end of the link.



Suppose that the packet length is $L = 8000$ bits, and that the link transmission rate along the link to router on the right is $R = 1$ Mbps.

Round your answer to two decimals after leading zeros



Question List

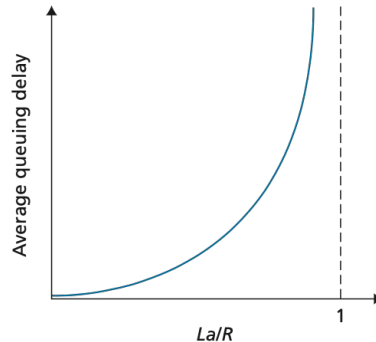
1. What is the transmission delay?
2. What is the maximum number of packets per second that can be transmitted by this link?

Solution

1. The transmission delay = $L/R = 8000 \text{ bits} / 1000000 \text{ bps} = 0.008 \text{ seconds}$
2. The number of packets that can be transmitted in a second into the link = $R / L = 1000000 \text{ bps} / 8000 \text{ bits} = 125 \text{ packets}$



Consider the queuing delay in a router buffer, where the packet experiences a delay as it waits to be transmitted onto the link. The length of the queuing delay of a specific packet will depend on the number of earlier-arriving packets that are queued and waiting for transmission onto the link. If the queue is empty and no other packet is currently being transmitted, then our packet's queuing delay will be zero. On the other hand, if the traffic is heavy and many other packets are also waiting to be transmitted, the queuing delay will be long.



Assume a constant transmission rate of $R = 1900000$ bps, a constant packet-length $L = 4700$ bits, and a is the average rate of packets/second. Traffic intensity $I = La/R$, and the queuing delay is calculated as $\frac{IL}{R(1 - I)}$ for $I < 1$.

Question List

1. In practice, does the queuing delay tend to vary a lot? Answer with Yes or No
2. Assuming that $a = 35$, what is the queuing delay? Give your answer in milliseconds (ms)
3. Assuming that $a = 77$, what is the queuing delay? Give your answer in milliseconds (ms)
4. Assuming the router's buffer is infinite, the queuing delay is 0.3815 ms, and 613 packets arrive. How many packets will be in the buffer 1 second later?
5. If the buffer has a maximum size of 627 packets, how many of the 613 packets would be dropped upon arrival from the previous question?



Solution

1. Yes, in practice, queuing delay can vary significantly. We use the above formulas as a way to give a rough estimate, but in a real-life scenario it is much more complicated.

$$I = \lambda a / R = 4700 * 35 / 1900000 = 0.865$$

$$2. \text{ Queuing Delay} = \frac{I \lambda}{R(1 - I)} * 1000 = \frac{0.865 * 4700}{(1900000 * (1 - 0.865))} * 1000 = 0.23 \text{ ms.}$$

$$I = \lambda a / R = 4700 * 77 / 1900000 = 0.19$$

$$3. \text{ Queuing Delay} = \frac{I \lambda}{R(1 - I)} * 1000 = \frac{0.19 * 4700}{(1900000 * (1 - 0.19))} * 1000 = 0.58 \text{ ms.}$$

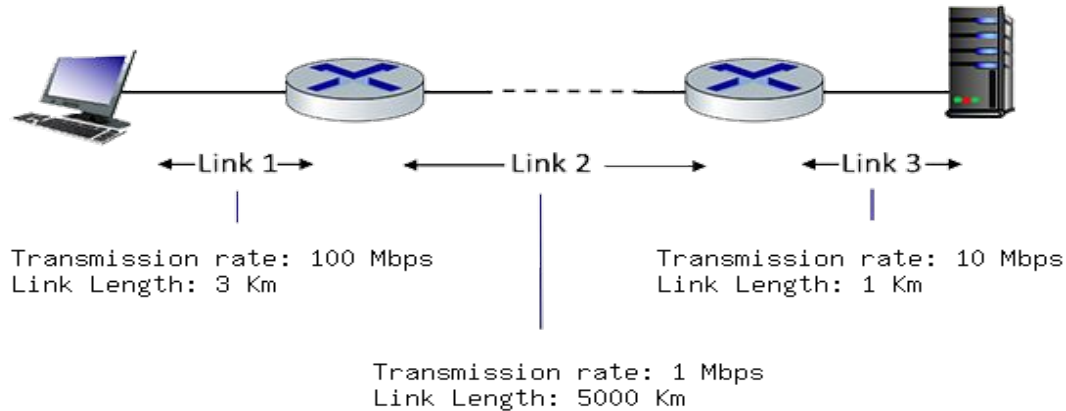
$$4. \text{ Packets left in buffer} = a - \text{floor}(1000 / \text{delay}) = 613 - \text{floor}(1000 / 0.3815) = 0 \text{ packets.}$$

$$5. \text{ Packets dropped} = \text{packets} - \text{buffer size} = 613 - 627 \text{ hence } 0 \text{ dropped packets.}$$



Computing end-end delay (transmission and propagation delay)

Consider the figure below, with three links, each with the specified transmission rate and link length.



Assume the length of a packet is 12000 bits. The speed of light propagation delay on each link is 3×10^8 m/sec

Round your answer to two decimals after leading zeros

Computing end-end delay (transmission and propagation delay)

Question List

1. What is the transmission delay of link 1?
2. What is the propagation delay of link 1?
3. What is the total delay of link 1?
4. What is the transmission delay of link 2?
5. What is the propagation delay of link 2?
6. What is the total delay of link 2?
7. What is the transmission delay of link 3?
8. What is the propagation delay of link 3?
9. What is the total delay of link 3?
10. What is the total delay?



Computing end-end delay (transmission and propagation delay)

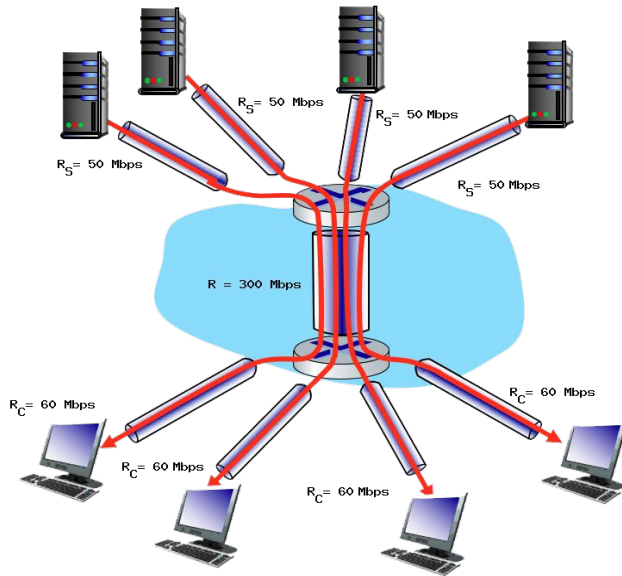
Solution

1. Link 1 transmission delay = $L/R = 12000 \text{ bits} / 100 \text{ Mbps} = 0.00012 \text{ seconds}$
2. Link 1 propagation delay = $d/s = (3 \text{ Km}) * 1000 / 3*10^8 \text{ m/sec} = 1.00\text{E-}5 \text{ seconds}$
3. Link 1 total delay = $d_t + d_p = 0.00012 \text{ seconds} + 1.00\text{E-}5 \text{ seconds} = 0.00013 \text{ seconds}$
4. Link 2 transmission delay = $L/R = 12000 \text{ bits} / 1 \text{ Mbps} = 0.012 \text{ seconds}$
5. Link 2 propagation delay = $d/s = (5000 \text{ Km}) * 1000 / 3*10^8 \text{ m/sec} = 0.017 \text{ seconds}$
6. Link 2 total delay = $d_t + d_p = 0.012 \text{ seconds} + 0.017 \text{ seconds} = 0.029 \text{ seconds}$
7. Link 3 transmission delay = $L/R = 12000 \text{ bits} / 10 \text{ Mbps} = 0.0012 \text{ seconds}$
8. Link 3 propagation delay = $d/s = (1 \text{ Km}) * 1000 / 3*10^8 \text{ m/sec} = 3.33\text{E-}6 \text{ seconds}$
9. Link 3 total delay = $d_t + d_p = 0.0012 \text{ seconds} + 3.33\text{E-}6 \text{ seconds} = 0.0012 \text{ seconds}$
10. The total delay = $d_{L1} + d_{L2} + d_{L3} = 0.00013 \text{ seconds} + 0.029 \text{ seconds} + 0.0012 \text{ seconds} = 0.03 \text{ seconds}$



End to End Throughput and Bottleneck Links

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of $R = 300$ Mbps. The four links from the servers to the shared link have a transmission capacity of $R_S = 50$ Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of $R_C = 60$ Mbps.



Question List

1. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?
2. Which link is the bottleneck link? Format as R_c , R_s , or R
3. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_s)? Answer as a decimal
4. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_c)? Answer as a decimal
5. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal



Solution

1. The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 50 Mbps
2. The bottleneck link is the link with the smallest capacity between R_s , R_c , and $R/4$. The bottleneck link is R_s .
3. The server's utilization = $R_{\text{bottleneck}} / R_s = 50 / 50 = 1$
4. The client's utilization = $R_{\text{bottleneck}} / R_c = 50 / 60 = 0.83$
5. The shared link's utilization = $R_{\text{bottleneck}} / (R / 4) = 50 / (300 / 4) = 0.67$





THANK YOU

Team Networks

Department of Computer Science and Engineering